

Exp 5: Introduction to Jenkins: What is Jenkins?, Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use

Jenkins is an open-source automation server that helps developers build, test, and deploy software automatically. It is widely used for **Continuous Integration (CI)** and **Continuous Delivery (CD)** in DevOps pipelines.

Key Features:

- Automates building, testing, and deploying code.
- Supports integration with tools like Git, Maven, Docker, and more.
- Highly extensible via plugins (over 1800 plugins available).
- Provides a web interface to configure jobs and monitor builds.
- Can be set up as a master-slave architecture for scaling.

Step 1: Install JDK 17/21 on GNU/Linux - Ubuntu

```
sudo apt update
sudo apt install openjdk-17-jdk -y
```

Step 2: Add Jenkins repository and key

```
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee \
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
```

Step 3: Add Jenkins repo to sources list

```
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

Step 4: Update package list

```
sudo apt update
```

Step 5: Install Jenkins

```
sudo apt install jenkins -y
```

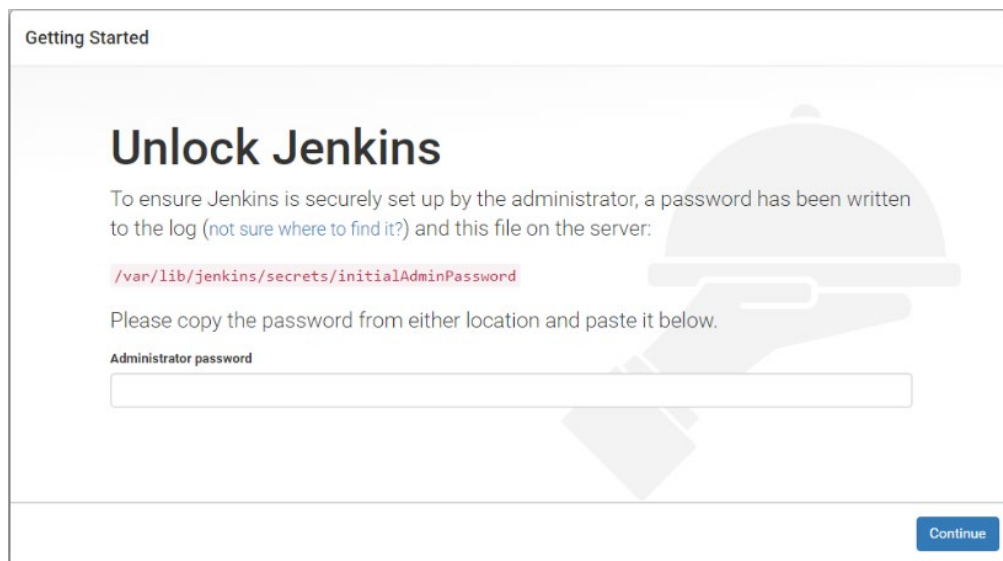
Step 6: Start and Enable Jenkins

```
sudo systemctl start jenkins
```

```
sudo systemctl enable Jenkins → (this enables Jenkins to start automatically at boot)
```

Step 7: To complete the installation setup, open a **web browser** and visit Jenkins on its default port (8080) using your server's domain name or IP address:

<http://localhost:8080>



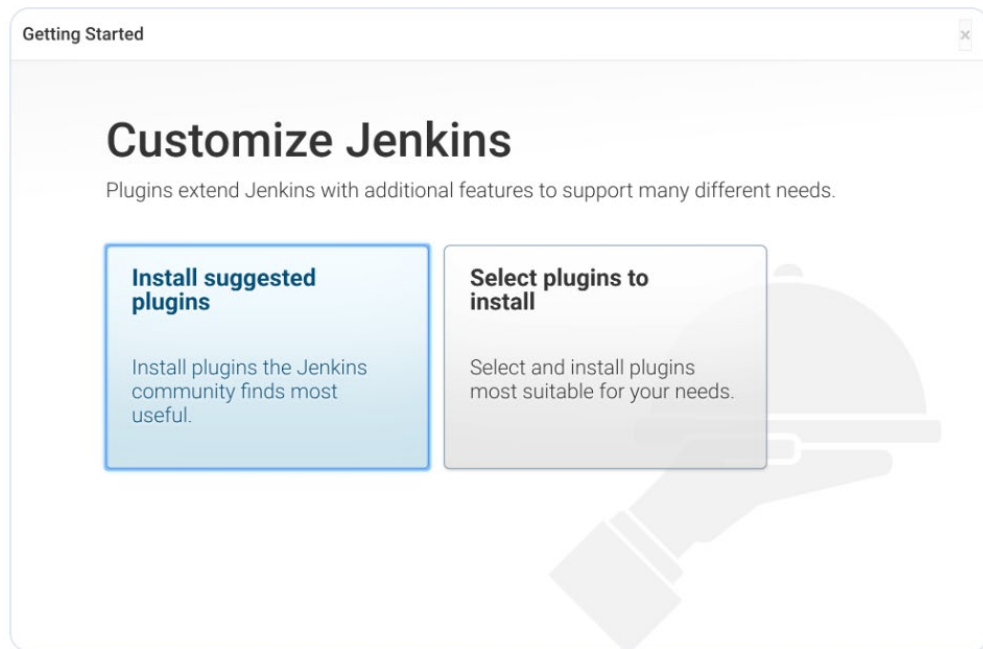
Unlock Jenkins screen, which displays the location of the initial password:

Step 8: **Get the admin password:** In the terminal window, use the **cat** command to display the password.

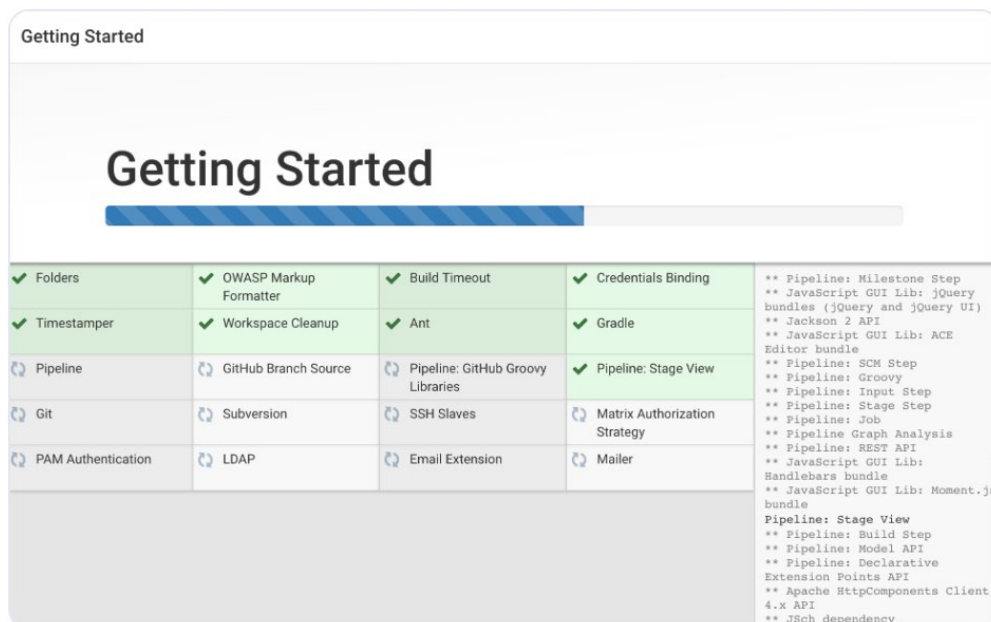
```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

- Copy the 32-character alphanumeric password from the terminal and paste it into the **Administrator password** field, then click **Continue**.

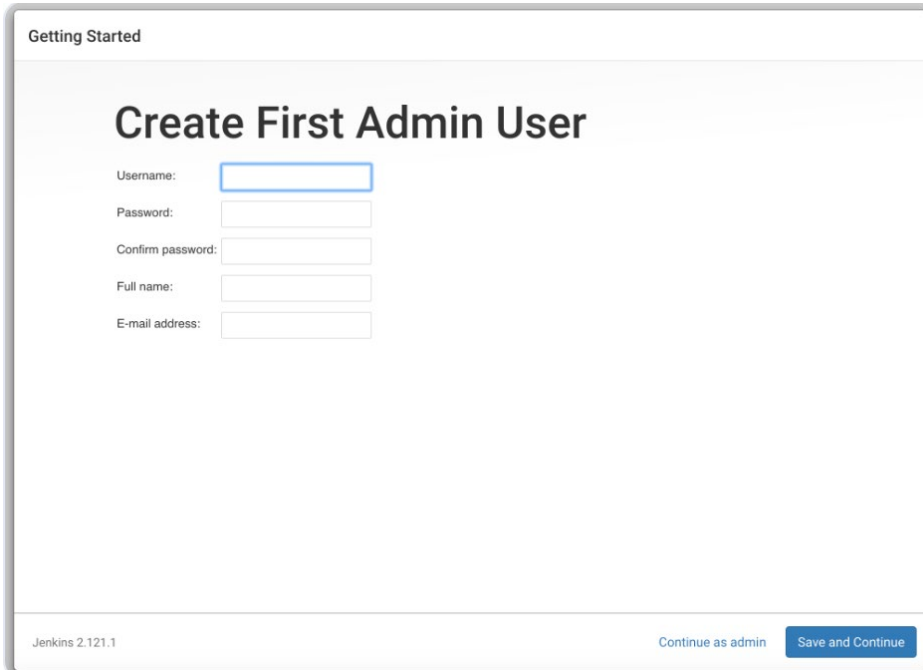
Step 9: The next screen presents the option of installing suggested plugins or selecting specific plugins:



Step 10: Click the **Install suggested plugins** option, which will immediately begin the installation process.



Step 11: When the installation is complete, you'll be prompted to set up the first administrative user - [Enter the name and password for your user](#)



The screenshot shows the 'Getting Started' page for Jenkins 2.121.1. The main heading is 'Create First Admin User'. Below the heading are five input fields: 'Username:', 'Password:', 'Confirm password:', 'Full name:', and 'E-mail address:'. At the bottom right, there are two buttons: 'Continue as admin' and 'Save and Continue'.

Getting Started

Create First Admin User

Username:

Password:

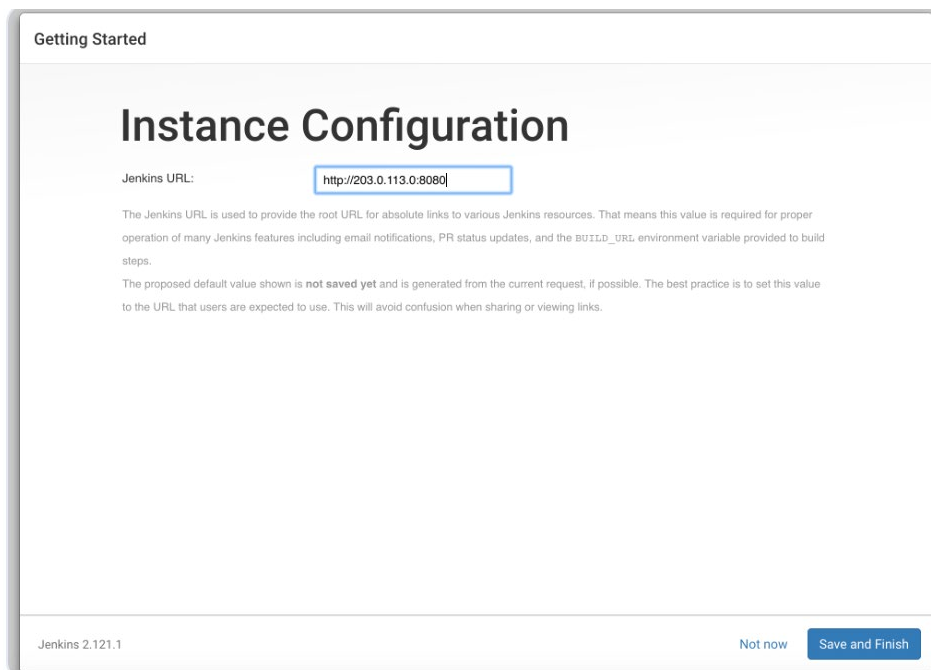
Confirm password:

Full name:

E-mail address:

Jenkins 2.121.1 [Continue as admin](#) [Save and Continue](#)

Step 12: You'll receive an **Instance Configuration** page that will ask you to confirm the preferred URL for your Jenkins instance. Confirm either the domain name for your server or your server's IP address:



The screenshot shows the 'Getting Started' page for Jenkins 2.121.1. The main heading is 'Instance Configuration'. Below the heading is a 'Jenkins URL:' label followed by a text input field containing 'http://203.0.113.0:8080/'. Below the input field is a paragraph of text explaining the Jenkins URL. At the bottom right, there are two buttons: 'Not now' and 'Save and Finish'.

Getting Started

Instance Configuration

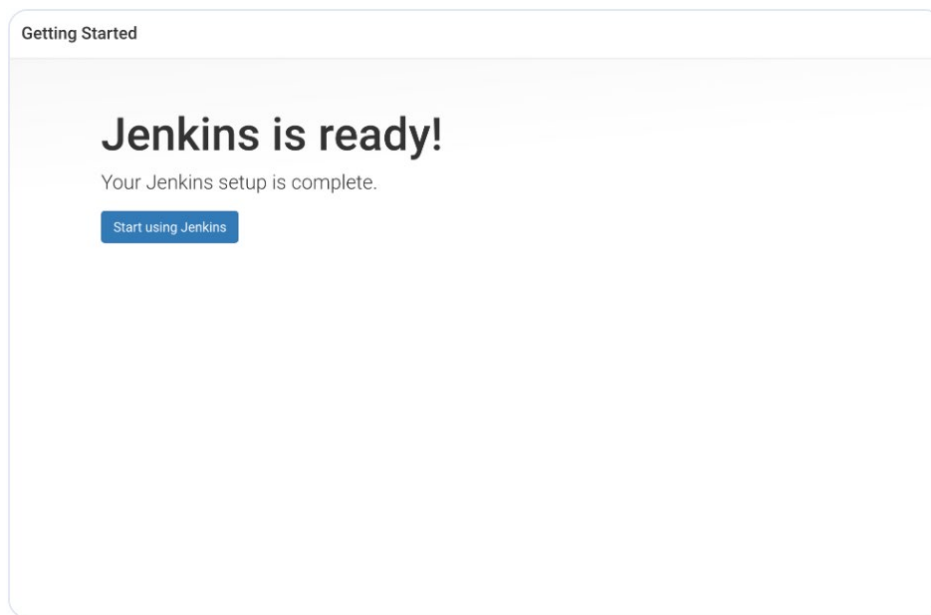
Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the `BUILD_URL` environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.121.1 [Not now](#) [Save and Finish](#)

Step 13: After confirming the appropriate information, click **Save and Finish**.



Step 14: Click **Start using Jenkins** to visit the main Jenkins dashboard:

